Extending a separately non-expanding rational function of two variables to the Cauchy reals

The Univalent Foundations Program

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Editorial note. The counted statement is the author's RC-binary-nonexpanding-extension (source lines 1404-1422); the proof environment begins on the next line after it, so the pairing is by adjacency.

Editorial note. The retained counter values (chapter 11, section 3, theorem offset 39) make the counted lemma print as Lemma 11.3.40, matching the first-edition numbering.

Editorial note. The author's proof works out the rational-against-limit constructor case and then states that the remaining conditions are the same computation; that sentence is retained as printed and is the author's own treatment, not an extraction omission.

Editorial note. The retained context is the paragraph that introduces the algebraic structure section, because it is where the author explains that the univariate extension gives negation and that addition needs the bivariate lemma counted here.

Editorial note. No retained line refers to a figure, bibliography entry, included file or non-ASCII character; the only reference outside the retained ranges is the univariate extension lemma declared above.

Editorial scope. One original result of the HoTT Book, counted once: the extension of a separately non-expanding rational function of two variables to the Cauchy reals.

Allowed standard prerequisites (editorial).

- The higher inductive-inductive Cauchy reals (R_C, closeness). R_C is generated by rat : Q -> R_C together with lim on Cauchy approximations, simultaneously with the family u close_eps v generated by four constructors relating rational points and limits; (R_C,close)-recursion produces f : R_C -> A from f(rat q), f(lim x), a separated family of relations on A and the four constructor conditions.. The retained proof applies that principle twice over, once with A the type of non-expanding maps R_C -> R_C (source lines 1425-1444) and once inside the base case through the univariate extension lemma.
- Non-expanding (Lipschitz with constant 1) maps R_C -> R_C. h : R_C -> R_C is non-expanding when u close_eps v implies h(u) close_eps h(v);

the proof forms the type A of such maps and the relation h ~_eps k := for all u, h(u) close_eps k(u) on it.. Defined and used at source lines 1426-1444; the whole currying trick of the proof is that the limit clause must produce an element of A, not merely a map.

- The univariate Lipschitz extension lemma. A Lipschitz f : Q -> R_C with constant L extends to f-bar : R_C -> R_C with the same constant and f-bar(rat q) = f(q) (book Lemma 11.3.15).. Used in the base case at source line 1448 to turn r |-> f(q,r) into an element of A; the lemma lies outside the retained span and is declared as a non-load-bearing navigation pointer.

Source reference. Original lemma 11.3.15: sources/hott-reals/pinned-original-source.tex, source lines 1073–1073. The retained lead-in and the base case of the retained proof cite the univariate Lipschitz extension lemma. It is a navigation pointer to an already available result of the same section: the counted lemma's own author proof is retained line for line, the cited lemma is not restated here, and no step of the counted proof is replaced by the pointer.

We first define the additive structure $(\mathbb{R}_\mathsf{c}, 0, +, -)$. Clearly, the additive unit element 0 is just $\mathsf{rat}(0)$, while the additive inverse $- : \mathbb{R}_\mathsf{c} \to \mathbb{R}_\mathsf{c}$ is obtained as the extension of the additive inverse $- : \mathbb{Q} \to \mathbb{Q}$, using original lemma 11.3.15 with Lipschitz constant 1. We have to work a bit harder for addition.

Lemma 11.3.40. *Suppose $f : \mathbb{Q} \times \mathbb{Q} \to \mathbb{Q}$ satisfies, for all $q, r, s : \mathbb{Q}$,*

$$|f(q,s) - f(r,s)| \leq |q - r| \qquad \text{and} \qquad |f(q,r) - f(q,s)| \leq |r - s|.$$

Then there is a function $\bar{f} : \mathbb{R}_\mathsf{c} \times \mathbb{R}_\mathsf{c} \to \mathbb{R}_\mathsf{c}$ such that $\bar{f}(\mathsf{rat}(q), \mathsf{rat}(r)) = f(q,r)$ for all $q, r : \mathbb{Q}$. Furthermore, for all $u, v, w : \mathbb{R}_\mathsf{c}$ and $q : \mathbb{Q}_+$,

$$u \sim_\epsilon v \Rightarrow \bar{f}(u,w) \sim_\epsilon \bar{f}(v,w) \quad \text{and} \quad v \sim_\epsilon w \Rightarrow \bar{f}(u,v) \sim_\epsilon \bar{f}(u,w).$$

Proof. We use $(\mathbb{R}_\mathsf{c}, \sim)$-recursion to construct the curried form of $\bar{f}$ as a map $\mathbb{R}_\mathsf{c} \to A$ where A is the space of non-expanding real-valued functions:

$$A :\equiv \{\, h : \mathbb{R}_\mathsf{c} \to \mathbb{R}_\mathsf{c} \mid \forall(\epsilon : \mathbb{Q}_+).\, \forall(u, v : \mathbb{R}_\mathsf{c}).\, u \sim_\epsilon v \Rightarrow h(u) \sim_\epsilon h(v) \,\}.$$

We shall also need a suitable $\frown_\epsilon$ on A, which we define as

$$(h \frown_\epsilon k) :\equiv \forall(u : \mathbb{R}_\mathsf{c}).\, h(u) \sim_\epsilon k(u).$$

Clearly, if $\forall(\epsilon : \mathbb{Q}_+).\, h \frown_\epsilon k$ then $h(u) = k(u)$ for all $u : \mathbb{R}_\mathsf{c}$, so $\frown$ is separated.

For the base case we define $\bar{f}(\mathsf{rat}(q)) : A$, where $q : \mathbb{Q}$, as the extension of the Lipschitz map $\lambda r.\, f(q,r)$ from $\mathbb{Q} \to \mathbb{Q}$ to $\mathbb{R}_\mathsf{c} \to \mathbb{R}_\mathsf{c}$, as constructed in original lemma 11.3.15 with Lipschitz constant 1. Next, for a Cauchy approximation x, we define $\bar{f}(\mathsf{lim}(x)) : \mathbb{R}_\mathsf{c} \to \mathbb{R}_\mathsf{c}$ as

$$\bar{f}(\mathsf{lim}(x))(v) :\equiv \mathsf{lim}(\lambda\epsilon.\, \bar{f}(x_\epsilon)(v)).$$

For this to be a valid definition, $\lambda\epsilon.\, \bar{f}(x_\epsilon)(v)$ should be a Cauchy approximation, so consider any $\delta, \epsilon : \mathbb{Q}$. Then by assumption $\bar{f}(x_\delta) \frown_{\delta+\epsilon} \bar{f}(x_\epsilon)$, hence

$\bar{f}(x_\delta)(v) \sim_{\delta+\epsilon} \bar{f}(x_\epsilon)(v)$. Furthermore, $\bar{f}(\mathsf{lim}(x))$ is non-expanding because $\bar{f}(x_\epsilon)$ is such by induction hypothesis. Indeed, if $u \sim_\epsilon v$ then, for all $\epsilon : \mathbb{Q}$,

$$\bar{f}(x_{\epsilon/3})(u) \sim_{\epsilon/3} \bar{f}(x_{\epsilon/3})(v),$$

therefore $\bar{f}(\mathsf{lim}(x))(u) \sim_\epsilon \bar{f}(\mathsf{lim}(x))(v)$ by the fourth constructor of $\sim$.

We still have to check four more conditions, let us illustrate just one. Suppose $\epsilon : \mathbb{Q}_+$ and for some $\delta : \mathbb{Q}_+$ we have $\mathsf{rat}(q) \sim_{\epsilon-\delta} y_\delta$ and $\bar{f}(\mathsf{rat}(q)) \frown_{\epsilon-\delta} \bar{f}(y_\delta)$. To show $\bar{f}(\mathsf{rat}(q)) \frown_\epsilon \bar{f}(\mathsf{lim}(y))$, consider any $v : \mathbb{R}_\mathsf{c}$ and observe that

$$\bar{f}(\mathsf{rat}(q))(v) \sim_{\epsilon-\delta} \bar{f}(y_\delta)(v).$$

Therefore, by the second constructor of $\sim$, we have $\bar{f}(\mathsf{rat}(q))(v) \sim_\epsilon \bar{f}(\mathsf{lim}(y))(v)$ as required. □